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RESEARCH ARTICLE

Fetal Ultrasound Monitoring Using MedSegDiff-V2–DUCKNet Ensemble Segmentation and Explainable Head Circumference Estimation

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ABSTRACT Fetal health monitoring is a vital component of prenatal care. Ultrasonography is the primary imaging technique for assessing fetal development and detecting abnormalities. In this work, an AI-enhanced ultrasound monitoring system has been developed that automates the segmentation and analysis of fetal anatomical structures. We used the Radboudumc HC18 dataset comprising 1,335 fetal head ultrasound images with segmentation masks, pixel-size metadata, and ground-truth HC/AC measurements. We collected a novel dataset (Aalok dataset) comprising 400 2D ultrasound scans from a local hospital in Dhaka, Bangladesh. The scans were obtained using a Samsung WS80A Elite system, which covers gestational ages of 18–38 weeks and includes precise manual annotations by radiologists. It consists of imaging, biometric measurements, and clinical annotations for complete fetal estimation. We applied multiple segmentation architectures, including U-NeXt, Attention U-Net, TransUNet, DUCK-Net, and MedSegDiff-V2. Among them, MedSegDiff-V2 and DUCK-Net achieved the highest segmentation performance with Dice scores of 0.9852 and recall of 0.9864 with ∼70 minutes training time, outperforming DUCK-Net, TransUNet and UNeXt. The proposed ensemble framework combines MedSegDiff-V2 and DUCK-Net through learnable and attention-based fusion mechanisms. The learnable ensemble achieved a Dice score of 0.9892 and a Recall of 0.9871. The proposed MedSegDiff-V2-Duck-Net ensemble technique demonstrated a strong correlation between predicted and actual HC values with a correlation of 0.9988. Grad-CAM++ explainable AI technique, has been applied to the ensemble model, visualizing the key ROI for clearer human understanding. The proposed system significantly reduces examination time and interobserver variability and increases diagnostic precision. The collected Aalok fetal head segmentation dataset is publicly available at the Aalok Dataset GitHub Repository: https://github.com/Safwan-Ul-Islam/Aalok_Dataset.

INDEX TERMS Automated segmentation, explainable AI, fetal ultrasound, head and abdominal circumference, MedSegDiff-V2, weighted ensemble.

I. INTRODUCTION

High-risk pregnancies resulting from fetal abnormalities are one of the leading causes of complications in maternal and fetal health [1]. The high risk of pregnancy is a threat to both the mother and the developing offspring, leading to various complications, including birth defects and developmental abnormalities. Fetal abnormalities should be effectively monitored and managed to reduce the risks associated with high-risk pregnancies. Ultrasound imaging technique plays a

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pivotal role in providing a safe method for tracking the health of the expectant mother and the development of the fetus [2]. Ultrasound imaging is a non-invasive procedure that creates images of internal organs and is extensively used in various medical scans, such as abdominal, pancreatic, urological, and hepatic scans [3]. Its prevalence is mostly seen in the maternal and neonatal sector. During an ultrasound scan, various fetal parameters are measured in detecting high-risk pregnancy and abnormalities, i.e., microcephaly, macrocephaly, and numerous congenital disorders [4]. High-risk pregnancies, particularly in developing countries like Bangladesh, are a growing concern, with an increasing number of cases projected due to rising rates of fetal abnormalities and limited access to prenatal care.

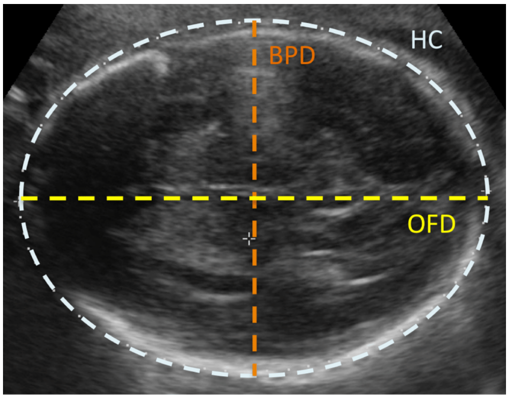

FIGURE 1. Fetal head biometric measurements: Head Circumference (HC), Biparietal Diameter (BPD), and OccipitoFrontal Diameter (OFD).

This work focuses on a deep learning models that automatically detect and segments fetal head circumference using 2D ultrasound images and using these outlines, the system will calculate important biometric measurements like Head Circumference, Biparietal Diameter, and Occipito-Frontal Diameter, depicted in Fig. 1, which are essential for tracking the baby's growth. Head circumference is measured around the outer perimeter of the fetal skull. The corresponding measurement is an indicator of fetal growth and neurological development. It plays an essential role in detecting abnormalities, including microcephaly and macrocephaly. Congenital disabilities are associated with genetic factors or intrauterine infections. The biparietal diameter represents the distance between the parietal bones of the fetal skull measured across its width. This measurement is most commonly used during mid-pregnancy to estimate gestational age and to track the overall growth of the fetus. The occipito-frontal diameter measures the length of the fetal skull from the frontal bone to the occipital bone. This technique provides insight into the longitudinal dimension of the head. When BPD and OFD are evaluated together. They offer a more complete understanding of cranial shape and development. The relationship between these two parameters is often expressed as the cephalic index [5].

This work implements automatic fetal head segmentation and circumference estimation techniques and presents the following key contributions:

- A novel dataset of fetal ultrasound images, labeled with accurately annotated head boundaries. The dataset comprises a diverse collection of 400 two-dimensional fetal ultrasound scans, obtained from Aalok Diagnostics Ltd, located in Dhaka, Bangladesh, and carefully reviewed and validated by domain experts.
- We investigate multiple fusion strategies for fetal head segmentation, including learnable weighted averaging and attention-based fusion, to combine the boundary-preserving strengths of MedSegDiff-V2 with the multi-scale feature learning capabilities of DUCK-Net. The proposed attention-based fusion framework adaptively emphasizes the most informative segmentation cues, resulting in improved boundary accuracy and clinically reliable head circumference estimation.
- An end-to-end automated fetal biometric measurement pipeline is developed that combines the proposed segmentation framework with contour-based circumference estimation. This framework enables direct extraction of fetal head circumference from ultrasound images without manual intervention.
- We demonstrate that the proposed hybrid framework achieves superior segmentation accuracy and circumference estimation performance compared with individual CNN-, transformer-, and diffusion-based segmentation models.

The novelty of this work lies in introducing a new expert-validated fetal head ultrasound dataset and a domain-adapted learnable hybrid ensemble that fuses diffusion-based boundary precision (MedSegDiff-V2) with multi-scale feature learning (DUCK-Net) to enable accurate automatic head circumference estimation.

The rest of the article is organized as follows: Section II discusses the major articles relevant to automated fetal biometric estimation from ultrasound images. Section III provides a detailed working overview of the proposed monitoring system, including the novel dataset, model architecture, segmentation process, circumference calculation and evaluation metrics. Section IV presents the key results obtained from the applied models and explainable AI techniques. The concluding remarks and possible future improvements are discussed in Section V.

II. RELATED WORKS

Conventional fetal health assessment, which involves manual interpretation of ultrasound images, requires highly trained and experienced sonographers and relies heavily on their subjective visual judgment. The manual diagnostic process is often time-inefficient, labor-intensive, prone to human error and involves variability between annotators. The availability of skilled diagnostic members in resource-limited regions, complicates timely and effective fetal evaluation.

Keerthi and Abirami [6] utilized a fetal ultrasound image dataset of 87 high-risk pregnant women with three categories to predict the risk of pregnancy failure during the second trimester. The authors performed data preprocessing that involved denoising, erosion, and dilation, and feature extraction using max pooling and bilinear interpolation. Segmentation results were refined by processing CNN outputs, ensuring accurate fetal measurements for categorical risk analysis. The ensemble CNN-TLFEM model outperformed the existing DL approaches, Deep-Lab-v3, with an mIoU of 0.90 and a Dice score of 0.96. Rathika et al. [7] developed a deep learning-based method to classify maternal-fetal ultrasound images for fetal organ identification and anomaly detection. The gray level co-occurrence matrix was used for feature extraction, producing 45 texture features. The authors improved classification accuracy by reducing characteristics using a hybrid optimization approach, achieving an accuracy of 98.07%.

Harikumar et al. [8] worked on classifying the image plane of fetal ultrasound using deep learning techniques and explainable AI. The hybrid CNN-U-Net model achieved training and validation accuracies of 99.58% and 86.35%, respectively. Kasera et al. [9] developed a deep learning-based approach to detect increased NT during the first trimester of pregnancy. The authors utilized a dataset of 560 standardized mid-sagittal ultrasound images, comprising 88 cases with thickened NT (> 3.5 mm). A U-Net architecture was used to segment both the fetus and the NT region, calculating the NT-to-fetus ratio for classification purposes. The U-Net model demonstrated a Dice score of 0.94 and an AUROC of 0.96.

Zeng et al. [10] applied a DAG V-Net deep learning method for automatic ultrasound image segmentation on the HC18 Challenge dataset. DAG V-Net integrated both attention mechanisms and deep supervision strategies into V-Net architecture. The DAG V-Net achieved a Dice similarity coefficient of 97.93%, a Hausdorff distance of 1.29 ± 0.79 mm, a HC difference of 0.09 ± 2.45 mm, and a HC absolute difference of 1.77 ± 1.69 mm. Yang et al. [11] developed the RDHCformer model, which combines the strengths of both transformers and CNNs. After evaluating the HC18 dataset, RDHCformer achieved strong accuracy with 1.72 MAE and 0.04 ME.

Coronado-Gutiérrez et al. [12] utilized a large dataset of 5,331 ultrasound images covering the main diagnostic planes. The proposed pipeline consisted of three main stages, which automatically classify brain planes, segment nine distinct brain structures, and extract measurements from the segmentation results. The system achieved an average classification accuracy of 98.6% for brain plane identification, an average pixel accuracy exceeding 96%, and a Jaccard index above 70% for structural segmentation. The automated biometric measurements showed an absolute error below 3.5% for the four major head metrics. Slightly higher errors were observed for the transcerebellar diameter (9%), the cavum septi pellucidi ratio (12%) and the Sylvian fissure operculization degree (26%).

Hayat and Aramvith [13] developed a weakly supervised medical image segmentation model for scribble-supervised learning. This model combines superpixel-guided graph attention, boundary-aware refinement, AFRB modules, and GAN-based learning to improve segmentation using sparse annotations. It was evaluated on the ACDC and MSCMRseg cardiac MRI datasets and achieved Dice scores of 0.902 and 0.871, respectively, outperforming methods such as ScribFormer and CycleMix.

Researchers effectively utilized various deep learning architectures, including advanced transformer models [14] and image preprocessing techniques for the automated analysis of fetal ultrasound images, as reviewed in relevant articles. Most studies relied on publicly available datasets, such as HC18, and evaluated the models using limited accuracy metrics without considering model or clinical applicability. Few studies have tackled challenges, including dataset diversity and variations in fetal positioning. These gaps have motivated us to design a more comprehensive evaluation framework that incorporates multiple performance metrics and a new, custom novel dataset. We enhanced our system with Explainable AI (XAI) tools, such as Grad-CAM++, that visualize critical regions that influence segmentation in automated fetal health assessments.

III. METHODOLOGY

A. DATASETS

1) HC18 DATASET

The image dataset used in this work is collected by Radboud University Medical Center, known as the HC18 dataset [15]. Fig. 2 shows representative fetal head ultrasound scans from the HC18 dataset, illustrating variations in fetal pose, gestational age, and image quality (e.g., speckle noise and low contrast).

2) CUSTOM AALOK DATASET

A significant contribution of this study is the development of a novel fetal ultrasound dataset collected from Aalok Diagnostic and Hospital, a specialized diagnostic center in Dhaka, Bangladesh. The dataset was curated under the clinical supervision of senior consultants in obstetrics and gynecology with extensive experience in prenatal imaging. It consists of 400 2D fetal ultrasound scans representing different gestational ages ranging from 18 to 38 weeks. The ultrasound images are 1024×768 pixels in dimension, with pixels ranging from 0.06 to 0.30 mm in size. These scans were recorded during routine antenatal check-ups using a Samsung WS80A Elite Ultrasound System. Fig. 3 shows representative fetal head ultrasound scans from the custom Aalok dataset.

Each ultrasound image was carefully reviewed and annotated by two experienced radiologists to ensure accurate delineation of the fetal head boundaries. The annotations were performed manually to serve as the ground truth for

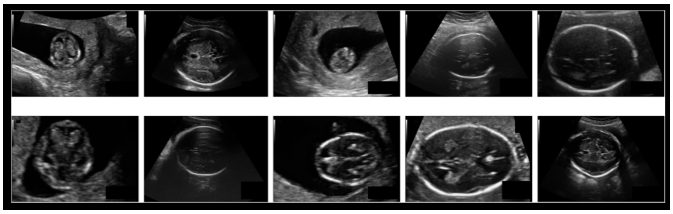

FIGURE 2. Sample fetal head ultrasound images from HC18 dataset.

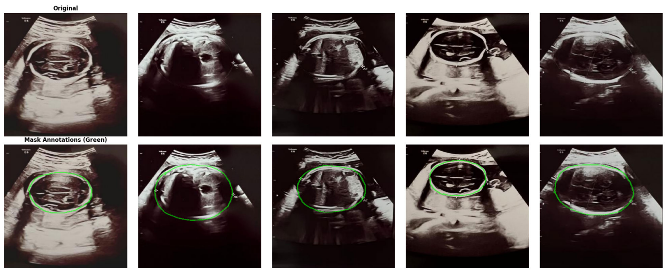

FIGURE 3. Sample fetal head ultrasound images from custom Aalok dataset.

training and validating deep learning segmentation models. The dataset includes diverse cases that capture variations in fetal orientation, image quality, and shadowing artifacts, making it representative of real-world clinical conditions.

TABLE 1. Dataset split distribution for training, validation, and testing.

Dataset	Train (70%)	Validation (15%)	Test (15%)	Total
HC18	935	200	200	1,335
Aalok	280	60	60	400
Combined	1,215	260	260	1,735

B. DATA PREPROCESSING AND AUGMENTATION

In this work, both datasets were preprocessed using the same preprocessing techniques and processes to ensure consistency and reliability across all of the scans. We employed several image preprocessing techniques to enhance the image and improve the segmentation results, which are described below.

- **Mask Refinement:** Although experienced radiologists manually delineated all fetal head masks, small internal holes occasionally appeared after mask resizing and file-format conversion. These artifacts did not reflect anatomical structures and could introduce inconsistencies during model training and contour-based circumference estimation. A hole-filling operation was applied to remove only isolated interior gaps while preserving the original external fetal head boundary. This post-processing step did not modify the expert-defined contour and was used solely to improve mask consistency.
- **Resizing:** Images and masks are resized to a fixed size of 256 × 256 while preserving the aspect ratio.
- **Augmentation (Training data only):** As we can see in Table 2, random flips, rotations, zooms, affine transformations, Gaussian noise, and contrast adjustments were applied. Fig. 4 illustrates augmented fetal head ultrasound images along with their segmentation masks and an overlay view.
- **Splitting:** Table 1 illustrates the distribution of the HC18 and Aalok datasets across the training, validation, and testing subsets using a 70:15:15 split ratio. The split was performed independently for HC18 and Aalok before augmentation, and the final training, validation, and test sets were created by concatenating the corresponding HC18 and Aalok partitions. Data augmentation was applied online during training, generating randomized transformed training samples at each epoch.

Fig. 5 presents the end-to-end pipeline starting from loading fetal ultrasound images and masks, applying intensity scaling/mask adjustments, and resizing all inputs to 256 × 256 and data augmentation (e.g., flips, rotations, Gaussian noise).

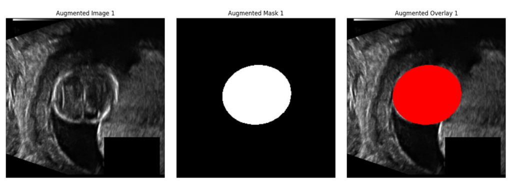

FIGURE 4. Representative augmented fetal ultrasound image and its corresponding fetal head segmentation mask from the Aalok dataset.

In this study, the applied model was trained using a Kaggle P100 GPU with 16 GB RAM to handle large datasets efficiently. The MONAI [16] library was used for medical imaging segmentation.

C. SEGMENTATION SYSTEM

Segmentation of the fetal head circumference from ultrasound images and calculating the fetal HC plays a vital role in monitoring fetal growth and identifying future health problems that might appear [3]. In most previous works,

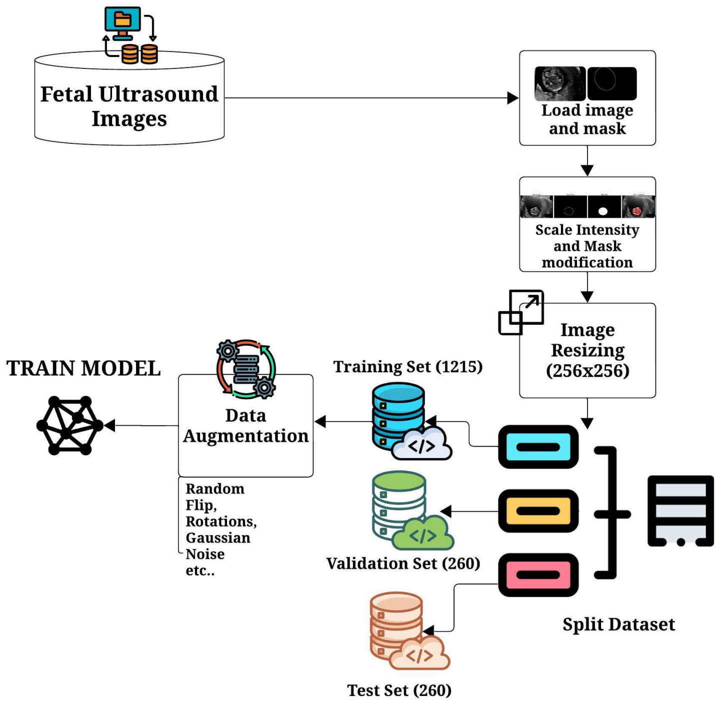

FIGURE 5. Dataset preprocessing steps for the proposed fetal ultrasound segmentation work.

TABLE 2. Augmentation techniques and their parameters.

Augmentation technique	Parameter
Random horizontal flip	probability: 0.5
Random vertical flip	probability: 0.5
Random Rotation	range:[-0.1 rad, $+0.1$ rad] $\approx [-5.7°, +5.7°]$
Random zoom	min zoom: 0.95, max zoom: 1.05, probability: 0.5
Random Gaussian noise	std: 0.01, probability: 0.3
Random contrast adjustment	gamma range: $[0.9, 1.1]$, probability: 0.3
Random affine transformation	rotate range: $\pi/30$ ($\sim 6°$), scale range: 0.05, translate range: 5, probability: 0.5

TABLE 3. Hyperparameters used to train applied models.

Hyperparameter	Value
Dropout Rate	0.2
Learning Rate	0.001
Weight Decay	0.0001
Focal Gamma	2.0
Focal Alpha	0.75
Boundary Loss	Enabled
Optimizer	AdamW
Scheduler	Cosine
Image Size	256×256
Batch Size	4
Random Seed	42
Batch Normalization	Instance
Gradient Clipping	1.0
Epochs	50
Steps per Epoch	100

the segmentation of the fetal head has been attempted using numerous convolutional network architectures, such as U-Net and various enhanced versions of U-Net, which are tailored for medical imaging [17] and semantic pixel-wise segmentation (SegNet) [18]. In our case, among all the implemented deep learning models, MedSegDiff-V2 and DUCK-Net demonstrated the most promising results in terms of segmentation accuracy and overall efficiency. Both models consistently achieved high Dice and IoU scores during validation, indicating superior capability in accurately capturing the fetal head boundaries from ultrasound images. Table 3 presents the hyperparameters used for training the applied models, including the 2D configuration, image size, channel progression, and stride settings.

1) MedSegDiff-V2

MedSegDiff-V2 [19] is an efficient approach to medical image segmentation. An important contribution of this study is the modification and optimization of the MedSegDiff-V2 architecture, which was specifically adapted for the characteristics of fetal ultrasound images. The architectural strength comes from the following key innovations.

a) Two-Track System: The core design is a Dual-Path Process. One track uses a time-conditioned U-Net to handle the actual denoising of the mask. The parallel track utilizes a CNN encoder, which extracts key semantic features from the original image.

b) Time Embedding: The time embedding makes the model aware of the step of the refinement process it is in. This allows the model to learn a step-by-step method for noise removal, making the process controlled and gradual.

c) Anchor-SA: The model uses Anchor Spatial Attention (Anchor-SA) to avoid getting distracted by the background.

It essentially lays down a learned Gaussian map that acts like a spotlight, forcing the model to focus its processing power precisely on relevant areas.

d) The Semantic Self-Former (SS-Former) for Consistency: The SS-Former [20] actively matches the noisy features of the evolving mask with the clean semantic context from the image. This framework maintains both structural and semantic integrity as the segmentation solidifies.

$$L_{\text{bce}} = \sum_i y_i \log o_i + (1 - y_i) \log(1 - o_i) \tag{1}$$

e) Hybrid Loss For training, a hybrid loss optimization is used that mixes Binary Cross-Entropy with Dice Loss. The binary cross-entropy loss penalizes pixel-wise classification errors and helps refine boundaries, and is calculated using (1).

$$\text{Dice Loss} = 1 - \frac{2 \sum_{i=1}^{N} p_i g_i + \epsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i + \epsilon} \tag{2}$$

where p_i and g_i denote predicted probability and ground-truth label for pixel i, respectively. The Dice loss measures the overlap between predicted and ground truth segmentation masks using (2). This work uses a hybrid loss function combining Binary Cross-Entropy loss and Dice loss. The hybrid loss is computed as the average of these two losses to ensure balanced optimization between pixel-wise classification performance and region-based segmentation quality.

This study presents one of the first domain-specific adaptations of such an architecture for fetal ultrasound data, integrating customized attention and conditioning mechanisms to address the unique challenges of low-contrast, noise-prone sonographic images. The proposed MedSegDiff-V2 architecture used for segmenting the fetal head from ultrasound images is illustrated in Fig. 6 below.

2) DUCK-NET

DUCK-Net [21] is an advanced U-Net model designed specifically for medical images. This model weaves an integrated deep encoder–decoder framework with efficient residual and skip connections. This process allows it to simultaneously grab the small, fine details and the big picture, the overall anatomical context. The applied DUCK-Net architecture used for segmenting the fetal head from ultrasound images has been illustrated in Fig. 7.

a) Multi-Stage Encoder–Decoder Layout: It uses the classic symmetric U-Net structure, an encoder that keeps compressing the image's spatial size while enriching its features (e.g., $F \times 2 \rightarrow F \times 4$), [22] followed by a decoder that rebuilds the high resolution mask.

b) DUCK Blocks: Every stage consists of a DUCK Block, similar to the network's specialized engine, which is a compact convolutional and residual unit that efficiently extracts features.

c) Lightweight Skip Connections: DUCK-Net manages its essential skip connections differently. Instead of simply concatenating features from the encoder to the decoder, it employs feature-wise addition. This step preserves all the crucial spatial information from the early layers and keeps the final segmentation boundaries sharp.

d) Refining during Rebuilding: The decoder upsamples the spatial resolution and the features get a clean-up. They are passed through residual blocks that stabilize the output.

e) Efficiency Boost with 1×1 Convs: The network uses 1×1 convolutions between major blocks. These layers compress the feature dimensions [21]. It retains the deep semantic knowledge without carrying redundant information.

f) The Bottleneck: The deepest layer is the semantic bottleneck. It encodes the highest-level global context (at $F \times 32$ channels), providing the network with a comprehensive structural understanding of the image before it begins the reconstruction phase.

3) ATTENTION U-NET

The Attention U-Net is a specialized U-Net architecture that incorporates attention gates into the skip connections [23]. These gates enhance the model's ability to focus on relevant regions of the fetal head while suppressing irrelevant areas, such as surrounding tissues or noise in ultrasound images. This approach is particularly valuable for fetal head segmentation, as ultrasound images often exhibit low contrast, noise, and unclear boundaries that are difficult to observe manually.

In this work, a five-level U-Net architecture is developed with channels (16, 32, 64, 128, 256). These channels help the model to progressively learn intricate features across four down-sampling steps defined by strides (2, 2, 2, 2). We applied instance normalization to handle the varying intensities that are typical in ultrasound images. We used ReLU activation to enhance feature extraction. The applied Attention U-Net model architecture is visualized in Fig. 8. Fig. 9 compares the input ultrasound image with the ground-truth mask and the model's predicted mask, shown as both a probability map and a binarized output.

4) MedSegDiff-V2 AND DUCK-NET ENSEMBLE

A significant contribution of this work is the combination of the MedSegDiff-V2 and DUCK-Net techniques. The segmentation outputs of DUCKNet and MedSegDiffV2 are fused using different ensemble strategies. It includes simple averaging, weighted averaging, and max fusion. For weighted averaging, both fixed and learnable weights are considered, where learnable weights are optimized through backpropagation and normalized using a SoftMax function. The resulting fused mask is treated as the final ensemble prediction for each input image. It is a two-stage training. The models are trained separately, the ensemble weights are trained (if learnable) using the frozen pre-trained models. Algorithm 1 describes an ensemble segmentation pipeline that combines predictions from pretrained MedSegDiff-V2 and DUCK-Net using multiple fusion strategies (average, fixed/learnable weighted average, and max).

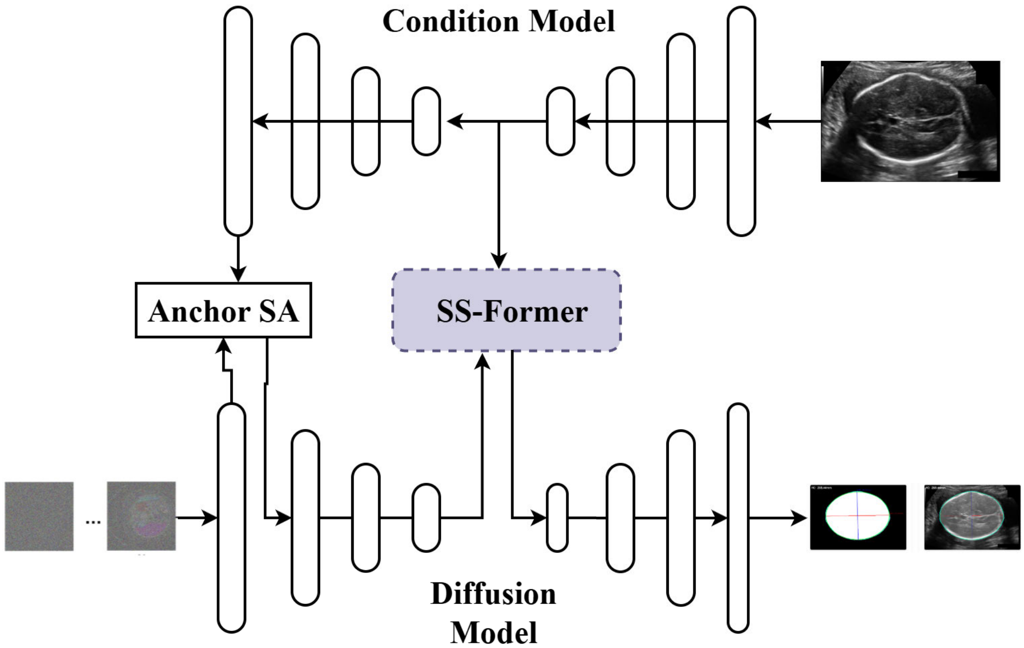

FIGURE 6. MedSegDiff-V2 architecture used for segmenting the fetal head from ultrasound images.

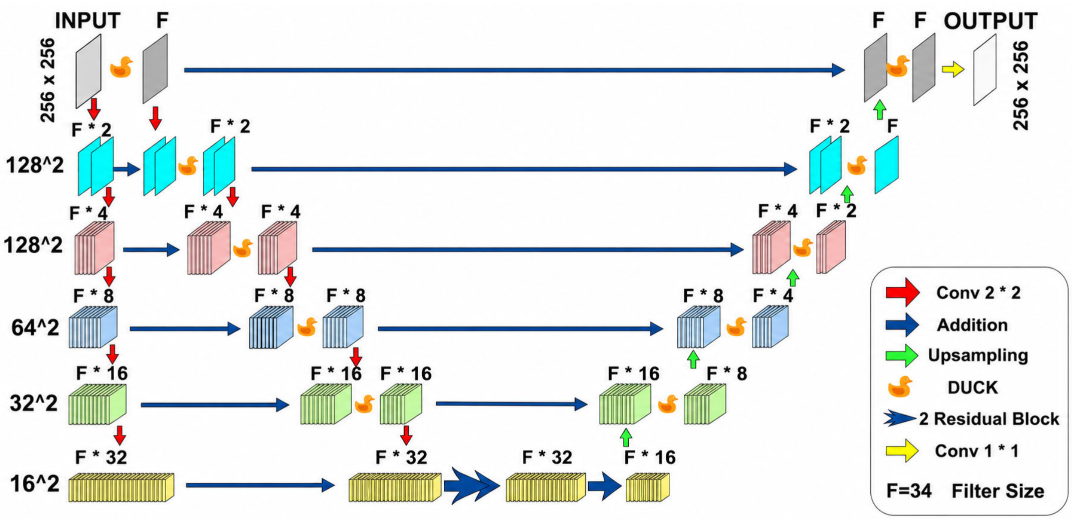

FIGURE 7. DUCK-Net architecture used for segmenting fetal head from ultrasound images.

Table 4 compares lighter diffusion-based segmentation alternatives with the adapted MedSegDiff-V2 framework used in this study. Although SegDiff and 3DKD-DiffuseNet have fewer parameters (∼23M and ∼21M, respectively), the adapted MedSegDiff-V2 is designed for 2D fetal ultrasound with Anchor-SA and FFT-based conditioning and uses only one inference step, despite its larger parameter size of approximately 68M.

D. CIRCUMFERENCE CALCULATION

The fetal head circumference (HC) was computed from the predicted binary mask following the standard clinical procedure used in commercial ultrasound systems. The segmentation output produced a binary mask of size $H \times W$, where pixel values correspond to the fetal skull region. From this mask, the largest connected contour was extracted using OpenCV to represent the fetal head boundary.

An ellipse was fitted to the extracted contour using a least-squares method, yielding the major and minor axes of the fitted ellipse. These axis lengths, initially measured in pixels, were converted to physical units (millimeters) using the pixel-to-millimeter spacing obtained from the ultrasound image metadata. The fetal head circumference was calculated using

TABLE 4. Gap analysis of diffusion-based segmentation methods for 2D fetal ultrasound.

Method	Params (M)	Modality	Speckle Robustness	Inference Steps	Key Limitation
SegDiff [24]	∼23	Natural / CT	Moderate	∼50 (DDIM)	Not tuned for ultrasound speckle
3DKD-DiffuseNet [25]	∼21	3D CT/MRI	Low	∼20 (DDIM)	Volumetric design; lossy on 2D
MedSegDiff-V2 (Proposed)	∼68	Fetal US (2D)	High (Anchor-SA + FFT)	1 (Algorithm 1)	Higher parameters; mitigated by single-shot inference

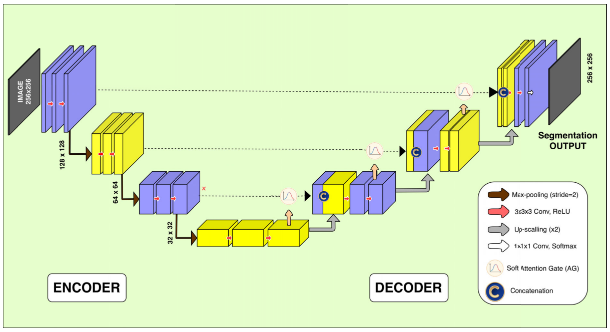

FIGURE 8. Model architecture for attention U-Net applied in this work.

Ramanujan's ellipse approximation as:

$$HC = \pi\left[3(m+n) - (3m+n)(m+3n)\right] \quad (3)$$

where m and n denote the semi-major and semi-minor axes of the fitted ellipse, respectively. This procedure produces HC values in millimeters and follows the same principle used in clinical fetal biometry systems.

E. OVERALL SYSTEM

GradCAM heatmaps were generated for sample validation images. For each sample, the image is passed through the model to obtain probability maps. A 0.5 threshold is applied to convert them into binary masks. Using the predicted masks, HC is computed by detecting contours and converting pixel measurements to physical units using a scaling factor, utilizing Eq. (3). Ground truth values were compared with the obtained predictions to validate the pipeline. Fig. 10 summarizes the overall methodology of the proposed fetal biometric estimation system.

For performance evaluation, we computed several error metrics to quantify differences between the predicted and ground-truth HC values. For qualitative assessment, we presented best-case and worst-case examples, showing the input images, predicted masks, and ground-truth masks side by side for comparison, illustrated in Fig. 10.

IV. RESULTS AND DISCUSSION

The results of the proposed fetal head segmentation system are discussed in this section.

Fig. 11 shows the qualitative fetal head segmentation result of the MedSegDiff-V2 + DUCK-Net model. Table 5 summarizes the quantitative performance metrics for different models. The best performing models for each metric have been given in bold letters.

From Table 5, it can be observed that the MedSegDiff-V2 achieved the highest overall performance across all metrics. It outperformed conventional CNN and transformer-based models, achieving a Dice score of 0.9852 and IoU of

Algorithm 1 Fusion-Based Ensemble of MedSegDiff-V2 and DUCK-Net

Require: Training data $\mathcal{D}_{train}$, Validation data $\mathcal{D}_{val}$, Pre-trained DUCKNet $\mathcal{M}_{duck}$, Pre-trained MedSegDiff-V2 $\mathcal{M}_{diff}$, Ensemble methods $\mathcal{E}$ = {average, weighted average, max}
Ensure: Optimized ensemble segmentation model

```
Phase 1: Individual Model Inference
for each image I ∈ D_val do
    S_duck^(I) = M_duck(I)
    Sample noisy mask x_t ~ N(0, I)
    Set T = 500 and inference timestep t = T/2 = 250
    Predict noise ε̂ = M_diff(x_t, I, t)
    Reconstruct clean mask:
        x̂_0 = (x_t − sqrt(1 − ᾱ_t) ε̂) / sqrt(ᾱ_t)
    Clamp x̂_0 to [−1, 1] and rescale to obtain S_diff^(I)
end for
Phase 2: Ensemble Fusion Strategies
for each ensemble method e ∈ E do
    for each image I ∈ D_val do
        if e = average then
            S_ensemble^(I) = (S_duck^(I) + S_diff^(I)) / 2
        else if e = weighted average then
            if learnable weights then
                Optimize w_duck, w_diff via backpropagation
                w = SoftMax([w_duck, w_diff])
                S_ensemble^(I) = w_1 · S_duck^(I) + w_2 · S_diff^(I)
            else
                S_ensemble^(I) = 0.5 · S_duck^(I) + 0.5 · S_diff^(I)
            end if
        else if e = max then
            S_ensemble^(I) = max(S_duck^(I), S_diff^(I))
        end if
    end for
end for
Phase 3: Performance Evaluation & Selection
for each ensemble method e ∈ E do
    Compute metrics on D_val:
        Dice = 2|S∩G| / (|S|+|G|),   IoU = |S∩G| / |S∪G|,   Accuracy
end for
Select optimal ensemble method e* with highest Dice score
return Trained ensemble model with method e*
```

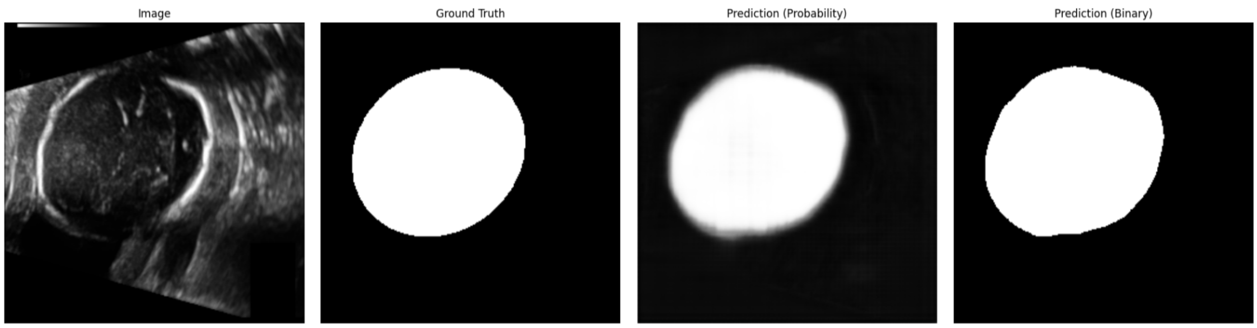

FIGURE 9. Example segmentation result for a fetal head circumference (HC) scan.

TABLE 5. Comprehensive performance comparison for fetal head segmentation on the combined dataset.

Metric	MedSeg-Diff-V2	DUCK-Net	Attention U-Net	UNeXt	TransUNet
Test Dice	**0.9852**	0.9682	0.9671	0.9615	0.9556
Test IoU	**0.9726**	0.9395	0.9365	0.9251	0.9128
Recall	**0.9864**	0.9699	0.9622	0.9630	0.9579
mAP	**0.961**	0.932	0.92	0.918	0.901
Training Time	70 min	175 min	41 min	40 min	33 min

0.9726 while requiring the training time of $\sim$ 70 minutes. Between baseline networks, DUCK-Net demonstrated strong stability and high Dice scores around 0.9682 but longer training times ($\sim$ 3 hours).

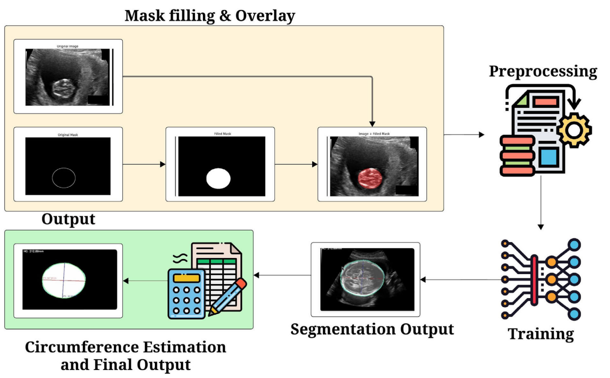

FIGURE 10. Working sequences of the proposed fetal head segmentation and head circumference estimation system.

TABLE 6. Computational complexity and pretraining status of applied models.

Model	Params (M)	GFLOPs	Inference (ms/img)	Pretrained Backbone	Size (GB)	Training Time
MedSegDiff-V2 (adapted)	68.4	38.4	127	None	9.4	70
DUCK-Net	25.1	12.7	18	None	5.1	175
Attention U-Net	34.9	18.2	22	None	4.7	41
TransUNet	96.7	26.4	42	ImageNet ViT-B/16	11.8	33
UNeXt	2.4	1.4	12	None	1.6	40
Ensemble (Learnable)	93.5	51.1	148	None	14.5	245
Ensemble (Attention)	93.6	51.4	152	None	14.5	55

Table 6 summarizes the computational complexity and pretraining status of the compared models, including parameter count, GFLOPs, and inference time per image. UNeXt was the lightest and fastest model with 2.4M parameters, 1.4 GFLOPs, and 12 ms/img inference time. The learnable ensemble required the highest computation with 93.5M parameters, 51.1 GFLOPs, and 148 ms/img, reflecting the trade-off between improved accuracy and computational cost.

Table 7 illustrates the five-fold cross-validation performance of the evaluated segmentation models on the combined HC18 and Aalok dataset ($n = 1,735$). The proposed ensemble of MedSegDiff-V2 + DUCK-Net with weighted average learnable technique achieved the best overall segmentation performance, obtaining a Dice score of 0.9880 ± 0.0013, IoU of 0.9742 ± 0.0033, Recall of 0.9857 ± 0.0011, and Accuracy of 0.9843 ± 0.0017. Among all methods, the Ensemble (Attention Fusion) produced the lowest Hausdorff Distance of 3.03 ± 0.18 pixels. The proposed learnable ensemble maintained consistently low standard deviations across all metrics.

To enhance segmentation stability, ensemble strategies combining MedSegDiff-V2 and DUCK-Net were explored. As shown in Table 8, the learnable weighted ensemble achieved the best overall segmentation performance, with a Test Dice of 0.9892 ± 0.011, IoU of 0.9787 ± 0.015, Accuracy of 0.9857, and Recall of 0.9871 ± 0.012. It obtained the lowest Hausdorff distance of 3.62 ± 0.84 px, indicating more accurate boundary localization. Compared with the fixed weighted average and simple average strategies, the learnable ensemble produced statistically significant improvements based on the Wilcoxon test ($p < 0.001$ and $p = 0.008$, respectively), although it required the longest training time of 55 minutes and slightly higher peak GPU memory usage of 14.7 GB. Fig. 12 shows the training and validation dices, with a clear improvement trend after around epoch 38. The best performance is Dice = 0.9892 (around epoch 40), and scores remain within the highlighted optimal region for several subsequent epochs, indicating stable high accuracy.

To avoid masking dataset-specific performance differences, we report the test results separately on the HC18-only and Aalok-only test subsets, in addition to the pooled combined-dataset results. Table 9 shows that the proposed learnable ensemble achieved the highest overall performance

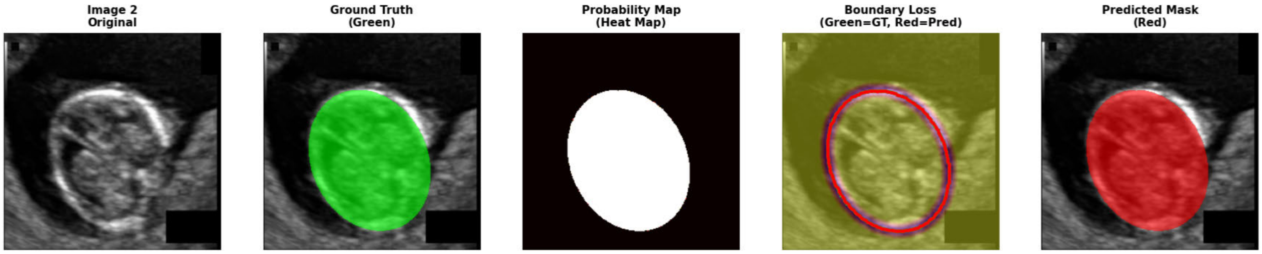

FIGURE 11. Visualization of best-case fetal head segmentation: Input ultrasound image, Ground truth mask (green, HC: 66.31 mm), Predicted probability map, boundary loss, Predicted binary mask (red, HC: 65.67 mm) with an absolute difference of 0.64 mm.

TABLE 7. Cross-validation results on the combined HC18 and Aalok dataset ($n = 1,735$). Values are reported as mean ± standard deviation across the five folds.

Model	Dice	IoU	Recall	Accuracy	Hausdorff (px)
DUCK-Net	0.9664 ± 0.0024	0.9368 ± 0.0036	0.9665 ± 0.0026	0.9619 ± 0.0010	4.34 ± 0.35
MedSegDiff-V2	0.9833 ± 0.0018	0.9696 ± 0.0048	0.9847 ± 0.0012	0.9724 ± 0.0018	3.93 ± 0.19
Attention U-Net	0.9640 ± 0.0025	0.9330 ± 0.0029	0.9613 ± 0.0013	0.9524 ± 0.0028	4.89 ± 0.37
UNeXt	0.9585 ± 0.0027	0.9198 ± 0.0061	0.9597 ± 0.0021	0.9526 ± 0.0036	5.07 ± 0.30
TransUNet	0.9534 ± 0.0025	0.9076 ± 0.0048	0.9548 ± 0.0050	0.9438 ± 0.0023	5.92 ± 0.68
Ensemble (W. Avg. Learnable)	$\mathbf{0.9880 \pm 0.0013}$	$\mathbf{0.9742 \pm 0.0033}$	$\mathbf{0.9857 \pm 0.0011}$	$\mathbf{0.9843 \pm 0.0017}$	3.90 ± 0.41
Ensemble (Attention Fusion)	0.9868 ± 0.0007	0.9741 ± 0.0009	0.9837 ± 0.0031	0.9827 ± 0.0017	$\mathbf{3.03 \pm 0.18}$

TABLE 8. Ensemble segmentation results of the MedSegDiff-V2 + DUCK-Net model with different fusion strategies.

Metric	Attention Fusion (Proposed)	Weighted Avg (Learnable)	Weighted Avg (Fixed)	Simple Average
Test Dice ± std	0.9868 ± 0.0007	0.9880 ± 0.0013	0.9727 ± 0.016	0.9789 ± 0.013
Bootstrap 95	[0.986, 0.988]	[0.987, 0.989]	[0.971, 0.975]	[0.977, 0.981]
Test IoU ± std	0.9741 ± 0.0009	0.9742 ± 0.0033	0.9469 ± 0.024	0.9587 ± 0.020
Accuracy ↑	0.9827	0.9843	0.9650	0.9758
Recall ± std	0.9837 ± 0.0031	0.9857 ± 0.0011	0.9758 ± 0.017	0.9713 ± 0.014
Hausdorff (px) ↓	$\mathbf{3.03 \pm 0.18}$	3.90 ± 0.41	4.97 ± 1.32	4.45 ± 1.18
Wilcoxon p (vs Attention Fusion)	–	0.012	< 0.001	< 0.001

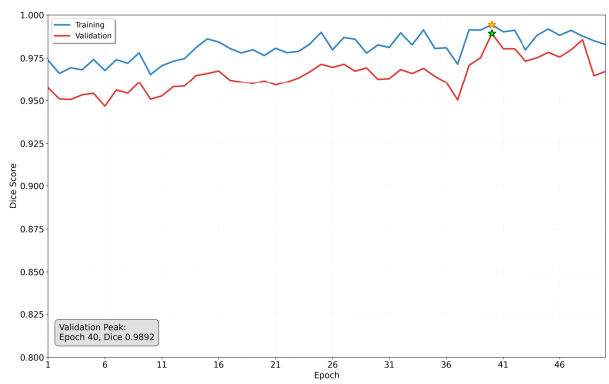

FIGURE 12. Epoch vs. training and validation dice for the MedSegDiff-V2+DUCK-Net ensemble model with learnable weighted average.

TABLE 9. Dataset-wise segmentation performance on HC18 and Aalok test sets.

Model	HC18 Dice	HC18 IoU	HC18 Recall	Aalok Dice	Aalok IoU	Aalok Recall
MedSegDiff-V2	0.9870	0.9745	0.9880	0.9790	0.9665	0.9810
DUCK-Net	0.9700	0.9425	0.9720	0.9620	0.9295	0.9630
Attention U-Net	0.9690	0.9395	0.9645	0.9605	0.9265	0.9540
TransUNet	0.9590	0.9170	0.9600	0.9445	0.8980	0.9510
UNeXt	0.9640	0.9285	0.9655	0.9530	0.9135	0.9550
Ensemble (Learnable)	0.9905	0.9800	0.9885	0.9850	0.9745	0.9830

TABLE 10. Cross-domain generalization performance (Train on HC18 dataset, Test on Aalok dataset).

Metric	MedSeg-Diff-V2	DUCK-Net	Attention U-Net	TransUNet	UNeXt
Validation Dice	**0.9815**	0.9677	0.9623	0.9621	0.9558
Validation IoU	0.9535	**0.9570**	0.9420	0.9383	0.9352
Test Dice	**0.9789**	0.9648	0.9555	0.9586	0.9597
Test IoU	0.9502	**0.9529**	0.9376	0.9242	0.9185
Recall	**0.9794**	0.9587	0.9521	0.9518	0.9440
mAP	**0.9678**	0.9513	0.9485	0.9185	0.9096
Training Time	45 min	75 min	32 min	25 min	30 min

on both datasets, obtaining Dice scores of 0.9905 on HC18 and 0.9850 on Aalok.

Table 10 evaluates cross-domain generalization by training on HC18 and testing on the Aalok dataset. MedSegDiff-V2 performs best overall, achieving Test Dice = 0.9789, Test IoU = 0.9502, Recall = 0.9794, and mAP = 0.9678. DUCK-Net is second-best (Test Dice = 0.9648) but requires much longer training (75 min) than UNeXt (30 min), Attention U-Net (32) and TransUNet (25 min), which show lower test performance.

Table 11 presents the reverse cross-domain experiment, where models were trained on Aalok and evaluated on HC18 to assess generalization in the opposite direction. The learnable ensemble achieved the best performance, with a

TABLE 11. Reverse cross-domain generalization results: training on Aalok and testing on HC18.

Model	Val Dice	Val IoU	Test Dice	Test IoU	Recall	Train Time
MedSegDiff-V2	0.9641	0.9298	0.9612	0.9287	0.9634	31 min
DUCK-Net	0.9512	0.9218	0.9487	0.9214	0.9521	58 min
Attention U-Net	0.9423	0.9115	0.9398	0.9102	0.9412	22 min
TransUNet	0.9388	0.9067	0.9356	0.9043	0.9388	18 min
UNeXt	0.9331	0.8991	0.9301	0.8978	0.9332	21 min
Ensemble (Learnable)	0.9714	0.9421	0.9683	0.9398	0.9712	44 min

Test Dice of 0.9683, Test IoU of 0.9398, and Recall of 0.9712, outperforming all individual models.

A. ABLATION STUDY

The ablation experiments highlight the performance gains obtained by systematically incorporating each architectural innovation within the MedSegDiff-V2 framework. As illustrated in Table 12, replacing Anchor-SA with U-SA results in a minor performance drop of 0.95% Dice, i.e., from 98.87% to 97.92%.

U-SA helps in general attention redistribution but lacks the explicit Gaussian localization bias that makes Anchor-SA more robust in handling ambiguous organ boundaries. The Anchor-SA variant remains the optimal configuration for MedSegDiff-V2.

Table 13 compares our approach with prior fetal ultrasound deep learning studies in terms of dataset type, network architecture, segmentation performance, and training time. Our proposed method (MedSegDiff-V2 + DUCK-Net with learnable weighted averaging on Private + HC18) achieves the best overall results (Accuracy = 0.985, Dice = 0.989, IoU = 0.951) while requiring the shortest training time (0.9 h) among the listed methods

B. HEAD CIRCUMFERENCE CALCULATION

In this work, fetal head boundaries were annotated and reviewed by two clinical experts, with an inter-observer Dice score of 0.91, indicating strong agreement between annotators. Any disagreements were resolved through consensus before finalizing the ground-truth masks. To validate the results of the proposed fetal head segmentation system through the head circumference calculation, we used various performance metrics. Table 14 provides a clinical sanity-check of head circumference estimation by reporting absolute errors, tolerance-band accuracy, outlier rate, Bland–Altman agreement, and bootstrap confidence intervals. The proposed MedSegDiff-V2 + DUCK-Net ensemble achieved the lowest MAE of 13.09 ± 8.42 mm, RMSE of 14.91 mm, and MAPE of 7.02%, outperforming MedSegDiff-V2, DUCK-Net, TransUNet, and UNeXt. The ensemble showed the highest agreement with ground truth, with 55.2% of cases within ±10 mm, 76.8% within ±20 mm, the lowest outlier rate of 2.8%, and the narrowest 95% limits of agreement of [−22.4, 18.1] mm.

Fig. 13 presents a comparative visualization of the MedSegDiff-V2 + DuckNet model, showing the relationship between the true and predicted head circumference values, along with the mean absolute error (MAE) and root mean squared error (RMSE). Most points cluster near the ideal line, indicating strong agreement between predictions and ground truth, with MAE = 13.09 mm and RMSE = 14.91 mm, and a strong correlation with $r = 0.9988$.

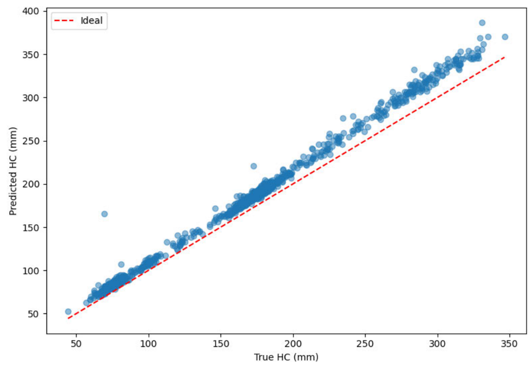

FIGURE 13. Scatter plot of true vs. predicted head circumference for the MedSegDiff-V2 + DuckNet model.

Fig. 14 analyzes the bias in the MedSegDiff-V2 + DuckNet model's head circumference predictions by plotting the difference between true HC and predicted HC in millimeters. The data points, distributed across actual HC values ranging from 50 to 350 mm, show a concentration. The most differences range between −50 and 50 mm, and occasional outliers up to −250 mm, suggesting minimum bias. The model's predictions generally align well with the actual values, which is consistent with its high correlation of 0.9988.

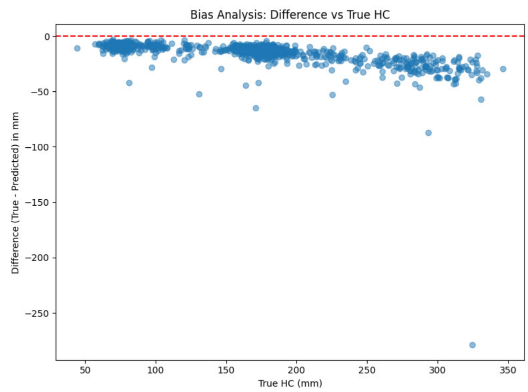

FIGURE 14. Bias analysis of head circumference prediction of the MedSegDiff-V2 + DuckNet model.

C. EXPLAINABLE AI

Although segmentation accuracy and contour agreement remain the primary evaluation criteria, we additionally employed Grad-CAM++ as a supplementary explainability tool to investigate whether the model attends to anatomically relevant fetal head regions. Unlike contour-based evaluation, Grad-CAM++ does not directly assess segmentation quality.

TABLE 12. Ablation study of MedSegDiff-V2 showing the incremental contribution of each architectural component. The replacement experiment demonstrates that Anchor Spatial Attention (Anchor-SA) achieves better segmentation quality than Uncertain Spatial Attention (U-SA), emphasizing its effectiveness in boundary-focused medical image segmentation.

Model Variant	Architectural Description	Dice (%)	IoU (%)
Diffusion-Only (Baseline)	Single-track time-conditioned U-Net denoiser without conditioning or attention modules; optimized with Binary Cross-Entropy (BCE) loss.	94.12	90.41
+ ConditionModel + Time Embedding	Adds CNN-based ConditionModel for semantic context and Time Embedding for progressive denoising refinement.	96.50	92.80
Full MedSegDiff-V2 (Anchor-SA)	Dual-track diffusion framework combining time-conditioned U-Net denoiser with semantic ConditionModel. Integrates Anchor Spatial Attention (Anchor-SA) and SS-Former transformer, trained with Hybrid (BCE + Dice) loss.	98.87	97.63
U-SA (Uncertain Spatial Attention) instead of Anchor-SA	Same as E3, but replaces Anchor-SA with Uncertain Spatial Attention (U-SA) to evaluate attention mechanism impact on localization precision.	97.92	96.34

TABLE 13. Comparison of existing fetal ultrasound deep learning studies with this work.

Ref.	Dataset	Network	Accuracy	Dice / IoU	Training Time
[6]	Private dataset	Ensemble CNN–TLFEM	0.954	Dice = 0.965	2.5 h
[7]	FETAL-PLANES DB	Custom CNN	0.938	–	3 h
[8]	Private + HC18	ResNet50 + Grad-CAM	0.942	–	2.8 h
[9]	Private dataset	U-Net	0.940	Dice = 0.940	2 h
[10]	HC18	DAG V-Net	–	Dice = 0.979	2.3 h
[11]	HC18	RDHCformer	0.970	Dice = 0.982	1.9 h
This work	Private + HC18	MedSegDiff-V2 (modified) + DuckNet with learnable weighted average	0.985	Dice = 0.989 / IoU = 0.951	0.9 h

TABLE 14. Clinical sanity-check results for head circumference estimation.

Metric	Attention Fusion	W. Avg. Ensemble	MedSegDiff-V2	DUCK-Net	TransUNet	UNeXt
MAE (mm) ↓	**12.55 ± 8.10**	13.09 ± 8.42	13.42 ± 9.11	14.92 ± 9.87	16.58 ± 10.23	17.03 ± 10.84
RMSE (mm) ↓	**14.35**	14.91	15.87	17.35	18.92	19.11
MAPE (%) ↓	**6.78**	7.02	7.85	8.10	8.74	8.92
Within ±5 mm (%)	**35.0**	33.5	31.0	28.4	24.6	25.1
Within ±10 mm (%)	**57.0**	55.2	52.1	47.8	43.9	43.1
Within ±20 mm (%)	**78.5**	76.8	73.5	69.1	65.4	63.7
Within ±5% (%)	**52.5**	51.2	47.8	42.5	38.7	37.4
Within ±10% (%)	**80.8**	79.5	76.4	71.8	67.2	66.0
Within ±15% (%)	**92.6**	91.8	89.5	85.6	82.4	81.7
Outlier > 50 mm (%) ↓	**2.3**	2.8	3.2	4.6	4.4	5.3
Pearson r ↑	**0.9990**	0.9988	0.9981	0.9965	0.9952	0.9956
Bland–Altman Bias (mm)	**-1.85**	-2.14	-2.87	-3.41	-4.12	-3.95
95% Limits of Agreement (mm)	**[-21.0, 17.3]**	[-22.4, 18.1]	[-24.5, 19.6]	[-27.2, 21.5]	[-30.1, 23.3]	[-31.0, 23.1]
Bootstrap 95% CI for MAE (mm)	**[11.40, 13.75]**	[11.83, 14.38]	[12.07, 14.89]	[13.54, 16.41]	[15.02, 18.21]	[15.47, 18.72]

This XAI technique provides a qualitative visualization of the image regions that most strongly influence the model's predictions. As shown in Fig. 15, the activation maps are concentrated around the fetal head and its boundaries, indicating that the model relies on clinically relevant structures during segmentation.

Fig. 16 presents failure-case Grad-CAM++ visualizations by comparing the input ultrasound image, ground-truth mask, predicted mask, and attention overlay. It shows that when segmentation errors occur, the model often attends to nearby high-contrast or misleading fetal structures, which helps explain why the predicted mask deviates from the true fetal head boundary.

D. LIMITATIONS

Although the proposed monitoring system demonstrated promising results, we observed a few limitations. The custom Aalok dataset used in this study was diverse but relatively small. It was collected primarily from a single hospital in Dhaka, Bangladesh, which limits the generalizability of the model to other populations and ultrasound machines. Future research could include data from multiple hospitals to enhance the model's generalizability. Although radiologists carefully reviewed all annotations, human judgment can vary, which may introduce some degree of bias in the ground truth data. Using larger datasets and consensus-based labeling approaches could reduce this variation. The explainability

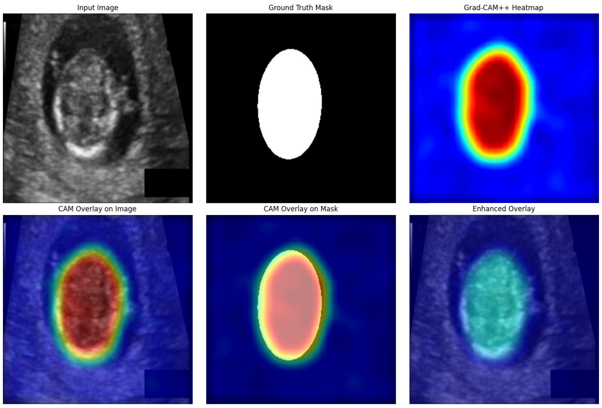

FIGURE 15. Grad-CAM++ visualization illustrating regions that contribute most strongly to the MedSegDiff-V2 segmentation prediction.

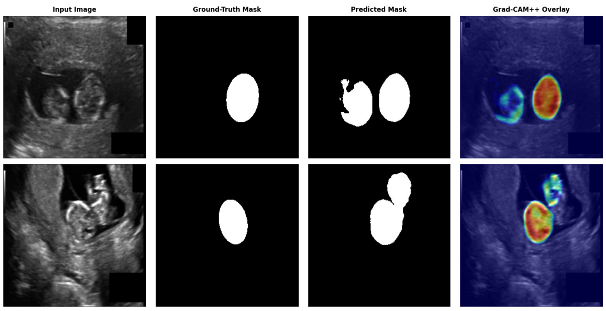

FIGURE 16. Failure-case Grad-CAM++ visualization for fetal head segmentation.

analysis implemented in this work has not been applied to validate segmentation decisions directly.

V. CONCLUSION

This study introduces an automated deep learning–based framework for fetal segmentation and HC estimation from 2D ultrasound images. Using a combined dataset comprising HC18 and data collected from Aalok Diagnostic Center, Dhaka, the proposed system was trained and evaluated across multiple architectures, including MedSegDiff-V2, DUCK-Net, Attention U-Net, TransUNet, and UNeXt. Among these, MedSegDiff-V2, a diffusion-based segmentation network modified with semantic conditioning and spatial attention, achieved a Dice coefficient of 0.985 and a recall of 0.986. The ensemble of MedSegDiff-V2 + DuckNet model with a learnable weighted average attained 0.989 and 0.987 Dice score and recall, respectively. The proposed ensemble technique demonstrated a strong correlation between predicted and actual HC values with a correlation of 0.9988. The use of explainable AI, as demonstrated by Grad-CAM++, showed interpretability by confirming that the models focused on clinically relevant head regions. The system was designed for offline deployment and to provide accessibility in low-resource or rural clinical environments.

In the future, a multimodal dataset can be developed that combines ultrasound images with maternal clinical parameters to improve model generalization. Comprehensive datasets from multiple hospitals, with annotations provided by additional radiologists, can be created. More complex tasks, including abdominal boundary separation, can be done employing 3D segmentation datasets. The proposed fetal segmentation and HC estimation system exhibits the potential to transform routine obstetric screening, supporting obstetricians in delivering accurate and efficient prenatal assessments worldwide. Future work may investigate inherently interpretable or attention-guided segmentation architectures that incorporate explainability within the model design itself.

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